
UNIT 6 INFLATION AND UNEMPLOYMENT *

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6.0 OBJECTIVES

After going through this unit you should be able to

- identify various types of unemployment;
- explain the concept of natural rate of unemployment;
- establish a relationship between unemployment and inflation;
- explain how the short-run Phillips curve shifts; and
- reconcile the difference in shape of the Phillips curve in short-run and long-run.

6.1 INTRODUCTION

In Units 7 and 8 of BECC 103 we gave some preliminary ideas of inflation – its definition, causes and effects. In this Unit we describe the relationship between inflation and unemployment. In the process, we discuss various types of unemployment and its measurement. Recall that classical economists believed in dichotomy of real and monetary variables. Thus, inflation being a monetary variable should not have any effect on a real variable such as unemployment. Keynesian economists, however, believed that change in monetary variables could affect real variables.

Inflation, as you know, is defined as a *persistent* rise or, a tendency towards persistent rise in the *general level of prices*. As and when there will be

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increase in the general price level, purchasing power of households decline. Increase in inflation is likely to reduce the value of national currency and it's vice versa. Although there are many types of inflation, it can be mainly classified into three types of inflation- demand pull (caused by increase in demand), cost push (cause by increase in cost of production) and built-in inflation (mainly caused by past events). The rate of inflation could vary from a low level to a very high level. As of 2021, Venezuela for example has an inflation rate of nearly 10,000 per cent per year.

Unemployment is a state in which healthy person fails to get employment at prevailing wage. It is caused by various factors that are concerned with demand and supply. There are broadly six types of unemployment- structural (arises due to change in structure of economy, frictional (arises due to time gap between changing jobs), cyclical (arises due to change in cycle of economy i.e. boom, recession, etc.), seasonal (arises due to change in season), disguised unemployment (it is also called hidden unemployment where marginal product of labour is very much close to zero) and under employment in which better qualified person end with low quality, low paid jobs because of excess supply of labour and lower availability of job opportunity.

6.2 TYPES OF UNEMPLOYMENT

‘Labour force’ as a concept includes all persons in the age group of 16 years to 64 years who are willing to work. Thus it includes both employed and unemployed persons. The persons not included in the labour force include those who are retired, too ill to work, keeping the house, or simply not looking for work.

‘Work force’ as a concept is somewhat narrower – it includes the employed persons only. Thus the difference between the labour force and the work force gives us the number of unemployed.

By employed persons we mean those who perform any paid work (thus homemakers are not included) and those who have jobs. On the other hand, the unemployed as a category includes people who are not employed but are actively looking for work. While considering unemployment we do not take into account those who are not in the labour force. We define unemployment rate as the number of unemployed divided by the total labour force. You should remember that the concept of unemployment implies ‘involuntary unemployment’. This concept implies that a person is willing to work at the prevailing wage rate, but cannot find work.

There are three types of unemployment, viz., frictional, structural and cyclical. We explain the differences below.

(i) Frictional unemployment: It takes place because people switch over from one job to another. In many cases the tenure of job gets over and workers remain unemployed till they get another job. In other cases workers migrate from one region to another in search of better jobs or opt to remain out of job for short time periods. Frictional unemployment takes place because in an economy with imperfect information, job search and matching is not smooth and there are frictions in the economy.

(ii) Structural unemployment: It results from the mismatch between supply and demand for different kinds of jobs. For example, in recent years, the number of engineers and management professionals looking for jobs in India has been much higher than available jobs. This has resulted in a number of persons with technical qualification opting for low qualification jobs. Structural unemployment takes place largely due to structural shifts in an economy and adjustments to such shifts take time. A large number of educational institutions in India have discontinued their engineering education programmes.

(iii) Cyclical unemployment: It arises due to fluctuations in aggregate demand, which is a part of business cycles. When aggregate demand declines, there is simultaneous decline in the demand for labour and consequent increase in unemployment. On the other hand, a general boom in the economy increases the demand for labour and unemployment decreases. Thus cyclical unemployment is pro-cyclical in nature.

Empirical data shows that the labour force in an economy is much less than the total population. Total labour force in India, according to certain sources, is about 50 crores compared to an estimated population of 138 crores in 2020. Persons above 65 years and children below 15 years of age however should not be taken into consideration while comparing the size of the labour force to total population. A relevant ratio in this context is the 'Labour Force Participation Rate (LFPR)'. It is defined as follows:

$$\text{LFPR} = \frac{\text{Size of the labour force}}{\text{Size of population in the age group of 16 – 64 years}}$$

The labour force participation rate (LFPR) varies across countries, and over time for the same country. If we take gender into account, there could be male labour force participation rate and female labour force participation rate. Usually, there is a gap between male LFPR and female LFPR. In India, for example, female LFPR is much lower compared to male LFPR. Further, there is a sharp decline in female LFPR in recent years. Such a decline could be due to cultural and structural issues.

The rate of unemployment u is defined as the ratio of unemployed persons to total labour force. The rate of unemployment varies across countries and for a country over time.

6.3 PHILLIPS CURVE

The Phillips curve, named after A W Phillips, describes the relationship between unemployment and inflation. In 1958 Phillips, then professor at London School of Economics, took time series data on the rate of unemployment and the rate of increase in nominal wage rate for the United Kingdom for the period 1861-1957 and attempted to establish a relationship. He took a simple linear equation of the following form:

$$\dot{w} = a - bu$$

where $\dot{w}$ is the rate of wage increase, a and b are constants and u is the rate of unemployment. Phillips found that there exists a stable and inverse relationship between $\dot{w}$ and u , with the implication that lower rate of unemployment is associated with higher rate of wage increase.

Subsequent to the publication of the results by Phillips, many economists followed suit and attempted similar exercises for other countries. Subsequently, it was established that there is a stable relationship between rising wage rate and rising price level. This led some economists to refine the simple equation estimated by Phillips and use of inflation (the rate of increase in prices) instead of wage rate increase. In many cases the scatter of plot of variables appeared to be a curve, convex to origin. As empirical studies reinforced the inverse relationship between the rate of inflation and the rate of unemployment the Phillips curve soon became an important tool of policy analysis.

The policy implication of such a result was astounding – an economy cannot have both low inflation and low unemployment simultaneously. In order to contain unemployment an economy has to tolerate a higher rate of wage increase and vice versa. Thus the Phillips curve justifies the discretionary stabilization policy of a government.

In Fig. 6.1 we depict a typical Phillips curve. Suppose the economy is operating at point A with inflation rate of π_1 and unemployment rate of u_1 . If the government wants to reduce the rate of inflation to u_2 , the economy has to tolerate a higher rate of inflation (π_2).

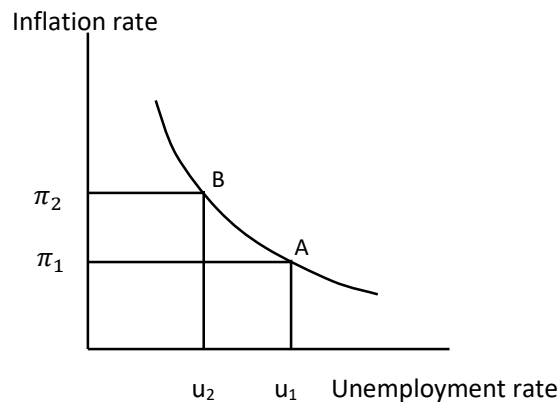


Fig. 6.1: Phillips Curve

During the 1960s and early 1970s the Phillips curve was considered to be an important tool of policy analysis. The prescription was simple and straight forward: During periods of high unemployment the government could follow an expansionary monetary policy which leaves more money in the hands of people. Such a policy may accelerate the rate of inflation while lowering unemployment. Conversely, during periods of high inflation the government could follow a contractionary economic policy so as to reduce inflation rate; the cost of such a policy however was supposed to be higher rate of unemployment. Thus, economists believed that there was a trade-off between inflation and unemployment. The government could choose any combination of inflation rate and unemployment rate depending upon the slope and position of the Phillips curve.

During the late 1970s and early 1980s, however, such a belief got shattered. The prescriptions of the Phillips curve did not work at all. Economies suffered from both high inflation and high unemployment. As unemployment increased, there was a lower level of output implying stagnation in economic growth. When governments tried to follow Keynesian policy prescription of higher government expenditure so as to increase aggregate demand, the rate of inflation accelerated. Thus, what most countries experienced was ‘stagflation’ – a combination of stagnation and inflation. The reason for stagflation was found to be supply shocks due the ‘oil crisis’ of 1973 and 1979 (refer to Unit 6). Stagflation prompted economists to explore further into the reasons for stagflation.

Check Your Progress 1

1. Write a brief note on the various types of unemployment.

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2. Explain how the Phillips curve could offer policy options before the government.

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3. Define the following concepts:
- a) Involuntary Unemployment
 - b) Natural Rate of Unemployment
 - c) Labour Force Participation Rate
 - d) Inflation-Unemployment Trade-Off

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6.4 NATURAL RATE OF UNEMPLOYMENT

You might have come across the term full employment, which implies that all workers in the economy are employed. Have you ever thought of such a situation? Can it be attained? When we say that an economy is operating at ‘full employment’ level, we do not mean that there is zero unemployment. Because of imperfections in markets, rigidities in wages and prices, and various frictions in the economy it is not possible to obtain zero unemployment.

For example, at any point of time, some workers are in the transition process from one job to another (frictional unemployment). Similarly, a fraction of workers cannot be employed because of mismatch between the skill they possess and the skill required (structural unemployment).

In view of the above, a new concept termed ‘natural rate of unemployment’ was introduced in the 1960s independently by Milton Friedman and Edmund Phelps. Natural rate of unemployment takes into account the frictions and imperfections in the economy and assumes that it is natural for an economy to have certain fraction of its labour force unemployed, at any point of time. We observe that any unemployment that is not natural could be due to business cycle, or policy related.

For empirical purposes natural rate of unemployment is the total of frictional unemployment and structural unemployment in an economy.

It varies across countries, and over time for the same country. For the US economy, for example, natural rate of unemployment is estimated to be between 3.5 per cent and 4.5 per cent. Many countries do not report any estimate of natural rate of unemployment.

The concept of natural rate of unemployment reshaped macroeconomic analysis in subsequent years. As we will see later in this Unit, expectations of economic agents (such as households, firms and government) about future economic environment play a major role in the shape and position of the Phillips curve.

6.5 EXPECTATION-AUGMENTED PHILLIPS CURVE

The Phillips curve discussed earlier could not explain stagflation in an economy. For explaining stagflation we need to bring in expectations into our analysis. In fact, Phillips curve given in Fig. 6.1 holds true if there is no change in expectations in the minds of people. In case people perceive that there is a change in expectations, then the Phillips curve will shift. Both adaptive expectations and rational expectations have important implications for Phillips curve.

6.5.1 Phillips Curve under Adaptive Expectations

You know from microeconomics that workers and employers take decisions regarding employment on the basis of real wage; not nominal wage. According to Friedman and Phelps, expectations do matter. Thus the 'expected real wage' should be looked into account for determining equilibrium output and wage rate.

Workers usually enter into a contract with the employer regarding their salary for certain time period. During contract period, salary cannot be re-negotiated; it can be changed only after the contract period is over. As the workers are aware of these conditions, they incorporate expected inflation into the contract. For example, if the workers expect that inflation rate would be 3 per cent in the coming year, they will negotiate the wage rate in such a manner that the real wage rate does not decline due to price increase.

For an expected inflation rate of π_1 per cent, suppose the Phillips curve is given by $SRPC_1$ (see Fig. 6.2). Suppose the economy is at point A. At this point the expected inflation rate is π_1 (say 3 per cent) and unemployment rate is at the natural rate u^* (say 6 per cent). At point A, the workers and firms expect an inflation rate of 3 per cent and they are getting it. Thus there is no pressure on the economy for a change.

There is a possibility of trade-off between inflation and unemployment along the curve $SRPC_1$. If there is higher inflation, then real wage will decline (because nominal wage cannot be increased due to existing contracts). Consequently, firms will employ more labour thereby leading to a decline in unemployment.

Suppose the government pursues an expansionist fiscal policy (government expenditure increases or tax rate decreases), which will boost aggregate demand. As a result, there is an increase in prices (means higher inflation rate). An expansionist monetary policy, such as increase in money supply or decrease in interest rate, will also have the same effect. It will lead to an increase in investment which will, to some extent, increase the demand for labour also. In either case, there is an increase in inflation rate to π_2 (say 6 per cent). The rate of unemployment reduces to u_2 (below the natural rate). It implies that more workers are employed, as a result of which output will be higher than potential output. In Fig. 6.2 we have shown this situation as point B.

Equilibrium at point B, however, is temporary. Workers very soon realize that there is an unexpected increase in inflation rate. In order to compensate for the price rise, workers will demand a higher wage rate. It will lead to a shift in the Phillips curve from $SRPC_1$ to $SRPC_2$. Equilibrium in the economy will be at point C in Fig. 6.2. Consequently, inflation will be at π_3 while unemployment will be at the natural rate, i.e., u^* .

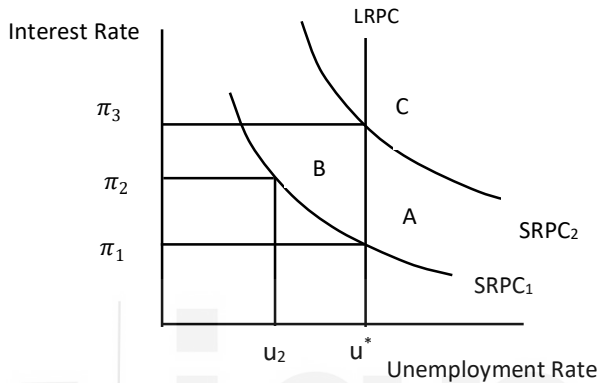


Fig. 6.2: Shift in Phillips Curve

Notice that unemployment in the economy is back at the natural rate (6 per cent). The inflation rate however is much higher (π_3). Thus the attempt of the government to reduce unemployment rate below the natural rate inflation rate, results in higher and higher inflation. In view of this, the natural rate of inflation is often termed as the ‘non-accelerating inflation rate of unemployment’ (NAIRU). It means that there is no acceleration in inflation if unemployment is maintained at this. Further, the term natural rate of unemployment indicates that it is inflexible, but social optimal. The term NAIRU, on the other hand, does not indicate any social optimality or desirability.

When unemployment is at the natural rate or NAIRU, there will be stability in the rate of inflation. When unemployment departs from the natural rate, there is acceleration or deceleration in inflation rate. Thus if actual unemployment is less than u^* , inflation will continue to accelerate – higher and higher in subsequent years. The concept of NAIRU and expectations formation explains the hyperinflation witnessed by many countries. Unless unemployment returns to its natural rate the inflation spiral will keep on accelerating.

The above analysis brings us to an important conclusion. Under adaptive expectations, the short run the Phillips curve is downward-sloping. In the long run however, it is vertical. In Fig. 6.2, the vertical line LRPC depicts the long run Phillips curve. Thus there is no trade-off between inflation and unemployment in the long run.

6.5.2 Phillips Curve under Rational Expectations

Under rational expectations economic agents such as firms and households are forward looking. They take into account all available information – past experience as well as present and future developments in the economy. There is no perfect foresight under rational expectations, but the errors cancel out on the whole.

An implication of the above is that actual inflation rate is equal to expected inflation rate. Thus workers and firms do not commit any error regarding wage rate during negotiations. Thus, there is no trade-off between inflation and unemployment under rational expectations. Unemployment rate in the economy is at the natural rate.

Suppose unemployment in the economy is at the natural rate. Firms and workers expect inflation to be at the rate of π_1^e . Suppose, the government pursues an expansionary policy as a result of which there is an increase in aggregate demand. Consequently, there is an increase in the rate inflation. If this policy was expected by the economic agents, they would have factored in the increase in inflation rate into their decision-making. If the policy is unexpected, then it would have the desired effect, that is, reduction in unemployment. This brings us to an important issue: how effective is government policy under rational expectations? If government policy is expected, it will not have any impact.

Check Your Progress 2

1. In 2019 the expected rate of inflation was 7 per cent while actual rate of inflation was 5 per cent. If $\lambda = 0.5$, find out the expected inflation rate for 2020.

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2. How do you reconcile the vertical long run Phillips curve with the downward sloping short run Phillips curve? Explain through a diagram.

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